DATA ANALYTICS PORTFOLIO PROJECT

Meta Ad Performance
Analysis Dashboard

An end-to-end Power BI project analyzing ad campaign performance, audience behavior and ROI across Meta platforms (Facebook & Instagram)

Prepared by

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Tools: Power BI • DAX • Data Modeling

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1. Executive Summary

This report documents the design, data model and findings behind the Meta Ad Performance Analysis Dashboard — a self-directed Power BI portfolio project that analyzes advertising performance across Facebook and Instagram. The dashboard consolidates campaign, ad-creative, audience and event-level data into a single interactive view covering reach, engagement, conversion efficiency and budget utilization.

Across 216K impressions and $2.5M in ad spend, the campaigns analyzed generated an 11.76% click-through rate and a 13.56% engagement rate — both well above typical industry benchmarks. However, only 0.61% of impressions converted into a purchase, revealing a significant drop-off between audience interest and completed transactions. The analysis translates this gap, along with audience, geographic, timing and creative-format patterns, into a set of concrete, prioritized recommendations.

> *The funnel's top is strong — creative and targeting are working. The bottleneck is conversion, and closing it is the single highest-leverage opportunity in this dataset.*

2. Project Objective

The goal of this project was to build an end-to-end BI solution — from raw event-level data to an executive-ready dashboard — that answers four core business questions:

- **Reach & Efficiency —** How far did the ads reach, and how efficiently was the budget converted into engagement and sales?
- **Audience Fit —** Which demographic and geographic segments respond best to the campaigns?
- **Timing —** When are users most likely to engage, and how should delivery be scheduled?
- **Creative Performance —** Which ad formats (Image, Video, Carousel, Story) deliver the best return?

The resulting dashboard is designed to support media planning, budget allocation and creative strategy decisions for a Meta advertising program.

3. Tools & Technologies

- **Power BI Desktop —** data modeling, DAX measures, and interactive report design.
- **DAX —** calculated KPIs including CTR, Engagement Rate, Conversion Rate and Purchase Rate.
- **Power Query —** data shaping, derived columns (day of week, time of day) and relationship building.
- **Star Schema Data Modeling —** one fact table and three dimension tables for scalable, efficient analysis.

4. Dataset & Data Architecture

The dataset simulates Meta's ad-platform event logs — covering campaigns, ad creatives, users and interaction events — and is modeled as a star schema with ad_events as the fact table and ads, campaigns and users as supporting dimensions.

Table	Role	Purpose
ad_events	**Fact Table**	Event-level log of every impression, click, share, comment and purchase; the source of all KPI calculations.
ads	**Dimension**	Ad-level metadata — platform, ad type/creative format, and targeting criteria (gender, age group, interests).
campaigns	**Dimension**	Campaign strategy, duration and budget; basis for cost-efficiency metrics such as CPC, CPM and ROAS.
users	**Dimension**	Demographic and location details of users who interact with ads; used for audience segmentation and targeting accuracy.

Relationships:

- ad_events → ads: links every interaction to the ad's platform, creative type and targeting.
- ads → campaigns: links each ad to its parent campaign's budget and timeframe.
- ad_events → users: links interaction events to the demographic profile of the user who performed them.

This structure lets a single event table drive every KPI (Impressions, Clicks, CTR, Conversion Rate, ROAS) while remaining filterable by platform, creative type, audience segment, geography and date — exactly the flexibility the dashboard's slicers expose.

5. Dashboard Overview

The dashboard is built as a single-page executive view with global filters for platform, campaign and target interest, and a dynamic measure selector that lets the KPI cards and trend charts be recomputed against different metrics on demand.

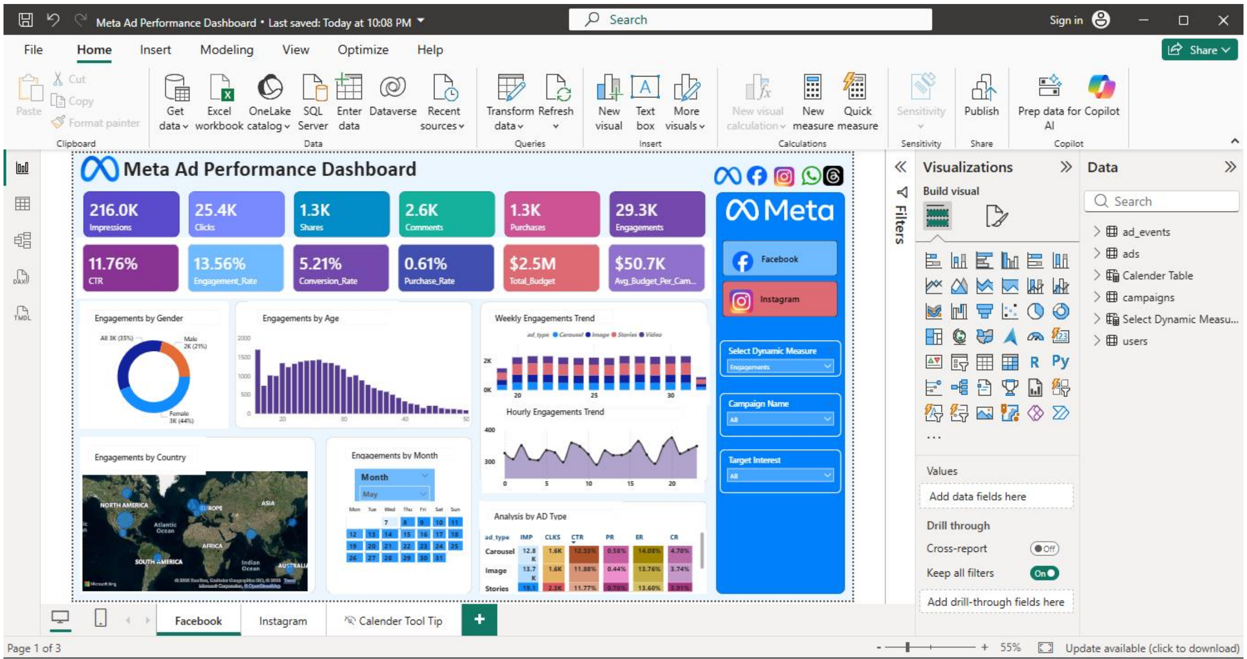

Figure 1 — Meta Ad Performance Dashboard (Power BI)

Layout highlights:

- **KPI strip —** Impressions, Clicks, Shares, Comments, Purchases, Engagements plus efficiency ratios (CTR, Engagement Rate, Conversion Rate, Purchase Rate) and budget figures.
- **Audience panels —** engagement by gender (donut) and by age (bar chart).
- **Trend panels —** weekly engagement (stacked bar by ad type), hourly engagement (line chart), and a calendar heat-view by day.
- **Geography —** a world map of engagement by country.
- **Creative performance —** a breakdown table of CTR, Purchase Rate, Conversion Rate and Engagement Rate by ad type.

6. Key Performance Indicators

The table below summarizes the headline metrics surfaced on the dashboard and what each one indicates about campaign health.

Metric	Value	What It Tells Us
Impressions	**216.0K**	Total number of times ads were displayed — strong overall reach.
Clicks	**25.4K**	Users who clicked through on an ad.
Engagements	**29.3K**	Sum of clicks, likes, shares and comments.
Shares / Comments	**1.3K / 2.6K**	Organic engagement signals beyond paid reach.
Purchases (Conversions)	**1.3K**	Confirmed customer acquisitions attributed to ads.
CTR (Click-Through Rate)	**11.76%**	Well above the ~1–2% industry benchmark — highly attractive creatives.
Engagement Rate	**13.56%**	Healthy audience resonance with ad content.
Conversion Rate	**5.21%**	Share of clicks that became purchases — solid, with room to improve.
Purchase Rate	**0.61%**	Share of impressions that became purchases — the weakest funnel stage.
Total Budget	**$2.5M**	Aggregate spend across all campaigns.
Avg. Budget / Campaign	**$50.7K**	Indicates a multi-campaign portfolio strategy.

7. Key Insights

7.1 Funnel Performance

CTR (11.76%) and Engagement Rate (13.56%) are both well above typical benchmarks, showing that ad creative, messaging and top-of-funnel targeting are highly effective — people are willing to click, like, share and comment. But Purchase Rate sits at just 0.61%, with only 1.3K purchases out of 216K impressions. This is a textbook case of strong awareness and interest paired with weak bottom-funnel conversion.

7.2 Audience: Gender & Age

- **Gender —** Female users account for 43% of engagement (13K) versus 22% for male users (6K), with the remainder unspecified. Female-skewed targeting shows a clear engagement edge.
- **Age —** Engagement peaks in the 20–30 age band, especially early 20s, and drops sharply after 35. The core audience is young adults.

> **Insight:**
> *Target ads toward women aged 18–30 for the strongest expected return.*

7.3 Geographic Distribution

The US, India, Brazil, Germany and the UK are the top-engaged markets. India and the US combine high volume with strong engagement, while Germany and the UK show smaller volume but higher purchasing power — better suited to premium offers than reach-driven campaigns.

7.4 Time-Based Trends

- **Weekly —** engagement is consistent week over week with no sharp drop-off, indicating the campaigns sustain attention over time rather than fatiguing quickly.
- **Hourly —** engagement peaks in the late afternoon and evening (roughly 15:00–20:00) and is lowest in the early morning (0:00–5:00).
- **Calendar —** certain dates show pronounced engagement spikes, consistent with the effect of specific promotions or launches.

7.5 Performance by Ad Format

Breaking the same KPIs out by creative type shows a clear ranking:

Ad Type	Impressions	Clicks	CTR	Purchase Rate	Conversion Rate	Engagement Rate
Carousel	48K	6K	11.7%	0.59%	5.1%	13.4%
Image	51K	6K	11.7%	0.57%	4.9%	13.5%
Stories	72K	8K	11.8%	0.65%	5.2%	13.6%
Video	46K	5K	11.9%	0.62%	5.2%	13.7%

- **Video** leads on CTR, Conversion Rate and Engagement Rate — the strongest all-round performer.
- **Stories** combines the highest impression volume with strong performance across every metric.
- **Image and Carousel** perform respectably but convert slightly less efficiently than Video and Stories.

8. Strategic Recommendations

Translating the insights above into a prioritized action plan:

1	**Fix the conversion funnel**	Purchase rate (0.61%) lags far behind CTR and engagement rate. Audit and optimize landing pages, checkout flow and offers to convert more of the interest already being generated.
2	**Refine audience targeting**	Concentrate spend on women aged 18–30, the segment already driving the strongest engagement, while testing incremental budget on the next best-performing segments.
3	**Prioritize creative formats**	Shift budget toward Video and Stories, which lead every performance metric, and treat Image/Carousel as supporting formats.
4	**Expand in high-value geographies**	Scale campaigns in India and the US for reach, and run premium, higher-AOV offers in Germany and the UK where purchasing power is stronger.
5	**Optimize delivery schedule**	Concentrate ad delivery in the afternoon–evening window (roughly 15:00–20:00) when engagement consistently peaks.
6	**Align spend with promotional peaks**	Calendar data shows engagement spikes around specific dates — plan budget increases and creative refreshes to coincide with these promotional windows.

9. Conclusion

This project demonstrates a complete BI workflow: modeling raw ad-event data into a star schema, building DAX measures for advertising KPIs, and designing an interactive Power BI dashboard that turns that data into decisions. The core finding — strong top-of-funnel performance undercut by weak purchase conversion — is the kind of insight a real advertising or growth team could act on immediately, whether by prioritizing landing page and checkout optimization, reallocating budget toward Video and Stories, or refining audience and geographic targeting.

From a portfolio standpoint, the project showcases skills across data modeling (star schema design), DAX (KPI and rate calculations), and dashboard/UX design (dynamic measures, drill-through, cross-filtering), alongside the analytical thinking needed to turn dashboard output into business recommendations.

10. About the Analyst

This project was built to apply core data analyst skills to a realistic, business-relevant scenario. The aim was to go through the full workflow a real analytics task demands: structuring raw ad-event data into a proper star schema, writing DAX measures for real advertising KPIs, and translating dashboard output into concrete business recommendations — not just charts for their own sake. It's part of a self-directed effort to build a portfolio that demonstrates job-ready data analyst skills through practical, end-to-end work.